Seminar 3: Solutions

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1 Try it Yourself: Add a Figure

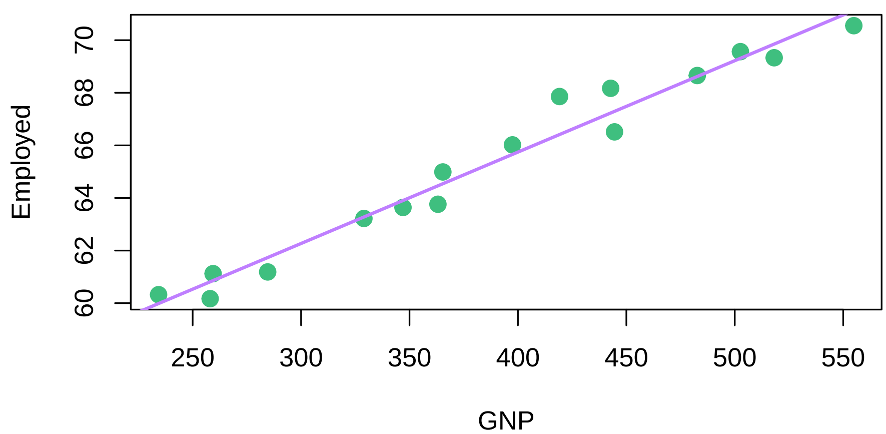

Figure 1: Scatterplot of Employed vs GNP for the longley dataset

Figure 1 shows a strong, positive, linear relationship between number of employed people and Gross National Product.

2 Try it Yourself: Add a Table

Table 1 shows a balanced study design in the ToothGrowth dataset.

3 Try it Yourself: Formatting Mathematics

For a discrete random variable, X, the expected value of X is calculated as:

Table 1: ToothGrowth factors.

supp	n
OJ	30
VC	30

$$E[X] = \frac{1}{n}\sum_{i=1}^{n} X_i$$

We can also calculate the variance of X, $\text{var}(X)$ as:

$$\text{var}(X) = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \text{E}[X])^2$$

The formatting here is incredibly pedantic. In the right situation, it's also incredibly important.

4 Try it Yourself: Formatting Environments

This is a solution.

5 Try it Yourself: Displaying Code You Don't Want to Run

```
calculate_mean <- function(x){
  n <- length(x)
  total <- 0
  for(i in 1:n){
    total <- total + x[i]
  }
  x_bar <- total / n
  return(x_bar)
}
calculate_mean(1:10)
```